



Exam 2 Review

November 10, 2025

Applied Comparative Genomics



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Assignment 5: Convolutional Neural Networks and Transformers

Assignment Date: Wednesday, October 29, 2024
Due Date: Friday, November 7 @ 11:59pm

Assignment Overview

In this assignment you will explore a couple of key aspects of convolutional neural networks such as self-attention and feature encoding as well as explore a pre-trained convolutional neural network for gene expression prediction. For this assignment, we will provide a Jupyter notebook with code for you to use and complete your assignment in.

For this assignment, you will create the environment as follows:

```
mamba create -n asn5 python=3.10 scikit-learn pytorch matplotlib pandas numpy jupyter seaborn kipoiseq logomaker
```

If installation fails due to "warning: libmamba - Problem type not implemented SOLVER_RULE_STRICT_REPO_PRIORITY", try running the following and then attempt the above command again:

```
conda config --set channel_priority flexible
```

Then activate the environment and install the following package using pip:

```
mamba activate asn5
```

```
pip install enformer-pytorch
```

If you have issues with creating the environment and installing the required packages, you can use [Google Colab](#)

You will need to add the following cells to the top of the notebook in Google Colab to install the dependencies:

```
!pip install kipoiseq==0.5.2
!pip install logomaker
!pip install enformer-pytorch
```

Because of the way Google Colab works, you will need to install these every time you reopen the notebook.

As a reminder, any questions about the assignment should be posted to [Piazza](#).

See the notebook here: [Assignment5.ipynb](#)

Packaging

The solutions to the above questions should be submitted as a single PDF document that includes your name, email address, and all relevant code, text, and figures (as needed). If you use ChatGPT for any of the code, also record the prompts used. Submit your solutions by uploading the PDF to [GradeScope](#), and remember to select where in your submission each question/subquestion is. The Entry Code is: GVXGV2.

If you submit after this time, you will use your late days. Remember, you are only allowed 4 late days for the entire semester!

Resources

- Jupyter notebooks: <https://jupyter.org/>

Computational Genomics: Applied Comparative Genomics
Project Presentations

Presentations will be a total of 9 minutes: 7 minutes for the presentation, followed by 2 minutes for questions. We will strictly keep to the schedule to ensure that all groups can present in class!

Schedule of Presentations

Slot	Day	Start	Team	Students	Title
1	11/19	3:00	Spatial Dynamics	Dhanav Handa, Arijit Mahesh Kulkarni	Learning Cellular Perturbation Dynamics in Foundation-Model Embedding Space via Neural Optimal Transport
2	11/19	3:09	V for Vogelsteins	Suki Ogihara	Cost-Effective Cancer Diagnostic via Aneuploidy dection with RealSeq5
3	11/19	3:18	Iso-Scope	Sai Dharmasena	Iso-Scope: Benchmarking Gene Isoform Inference Methods for Single-Cell Data
4	11/19	3:27	MTeam	Malavika Nair	Assessing Data Completeness in Variant Pathogenicity Prediction
5	11/19	3:36	Team Causality	Ziwei Jiang, Zihan Zhou	Causal Representation Learning for Gene Expression
6	11/19	3:45	Relish	Reena Assassa, Lillianna Gund, Shaun Ku	Comparing Genomic LLM Variant Pathogenicity Predictions with Established In-Silico Scores
7	11/19	3:54	Kmer-Bin matrix	Zhuolun (Julian) Du	A Probabilistic Approach to Hi-C Sequence Localization
8	11/19	4:03	Team KMers	Emily Sanchez Ocotlan, Maria Valverde-Cervantes, Kinverli Garcia Osorio	GenoViz: An Interactive Dashboard for Comparative Visualization of Personal Genomic Data
1	12/1	3:00	Team Zadorozhny	Nikola Zadorozhny	Fine-Tuning DNA Foundation Models for Variant Effect Prediction
2	12/1	3:09	AML Detection LLM	Charlotte Cheung, Claire Cui	Fine-tuning a Biomedical Language Model for Leukemia Subtype Prediction from Gene Expression Data
3	12/1	3:18	PrecisionMut	Rongrong Yan	Benchmarking Non-coding Mutation Impact Prediction in Colorectal Cancers Using Multi-Method Functional Scores
4	12/1	3:27	Team RNN	Alex Dong, Melody Jin, Katie Hong	Identification of Bacterial Promoters
5	12/1	3:36	Yolks to Folks	Megan Miller	Searching for Genetic Signatures and Constraints of Viviparity
6	12/1	3:45	Team Chauhan	Akshat Chauhan	Residual Variational Autoencoder for De Novo Design of Tumor-Specific Promoters
7	12/1	3:54	Julie X and Jason Z	Julie Xian, Jason Zavras	Evolutionary origin of Human Disease Genes: A Comparative Genomics Approach
8	12/1	4:03	LettUCE Grow	Xinyi Chen	Age vs Stress-induced inflammation: Universal Cell Embedding (UCE)-Based Comparative Profiling
1	12/3	3:00	Team Dark Magic	Ezra Greenberg	Analysis of MAGE Dataset for Microbial Contamination
2	12/3	3:09	Multimiomics	Matthew Wong	Integrated analysis of Multimiomics dataset for colon cancer patients; unravelling key prognostic biomarkers
3	12/3	3:18	MonoMarker	Fuchen Li, Huanchang Tu	Benchmarking single-cell trajectory inference under varying sequencing coverage using Monocle3
4	12/3	3:27	MetaGen	Charissa Luk, Rachael Pei, Jessie Huh	Predicting Type 2 Diabetes from Gut Microbiome Composition Through Classifier Benchmarking and Machine Learning
5	12/3	3:36	The Pink Helix	Lauren Ji, Sofia Floody	Investigating the Rate of Heterozygosity and Breast Cancer Among Global Populations
6	12/3	3:45	STalign2	Iris Kwon, Smriti Grounder	Improving STalign Accuracy by Incorporating Gene Expression
7	12/3	3:54	Chroma-Chrunchers	Martin Hilser	How Little Detail is Enough? Evaluating Chromatin-State Annotations Robustness Under Low-Coverage Sequencing

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<https://github.com/schatzlab/appliedgenomics2025/blob/main/project/presentations.md>

Computational Genomics: Applied Comparative Genomics
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Let me know ASAP on Piazza if you have a *major* conflict

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Recommended outline for your talk (30 seconds to 1 minute per slide):

1. Title Slide: Who are you, title, date
2. Intro: Whats the big idea???
3. Data: What data are you looking at?
4. Methods 1: What did you try?
5. Methods 2: What is the key idea?
6. Results 1: What did you see!
7. Results 2: How does it compare to other methods/data/ideas?
8. Discussion: What did you learn from this study? What are the next steps
9. Acknowledgements: Who helped you along the way?
10. Thank you!

I strongly *discourage* you from trying to give a live demo as they are too unpredictable for a short talk. If you have new software you want to show, use a "cooking show" approach, where you have screen shots of the important steps.

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What to study?

Closed book / no electronics in class exam [75 min]:

- One sheet of paper for notes (double sided)
- Recommend powerpoint / gslides / keynote but hand-written is okay
- Bring a pen / pencil!

1. Homework assignments!

- Make sure you understand the key concepts, algorithms, data structures, and methods involved
- Don't worry about running software or writing code, but you should know what are the major software tools and how they work

2. Lectures [with videos on canvas!]

- Make sure you understand the key concepts, algorithms, data structures, and methods involved :)
- No “complicated” numerical computing, but you should be able to sketch a distribution, approximate values, & other simple calculations

3. Readings

- These are to provide more details and provide additional perspectives
- You are not responsible for plots/content only presented in the readings
- Of course, you are free to read/watch/discuss additional resource

Questions will focus on material since exam 1 (HW4-5), but will include key topics from the first half of class

Exam Topics

Genomics

- Genomics Technologies
 - Illumina, PacBio, Nanopore
- Coverage, Kmers, Motifs
- Genome Assembly: GRCh38, T2T
- Whole Genome Alignment, Dotplot
- Read mapping
- Variant Identification
- Gene Finding
- Genome Annotation
- BLAST
- RNA-seq, Volcano Plots
- Single-cell analysis
- Methyl-seq, Chip-Seq, Hi-C

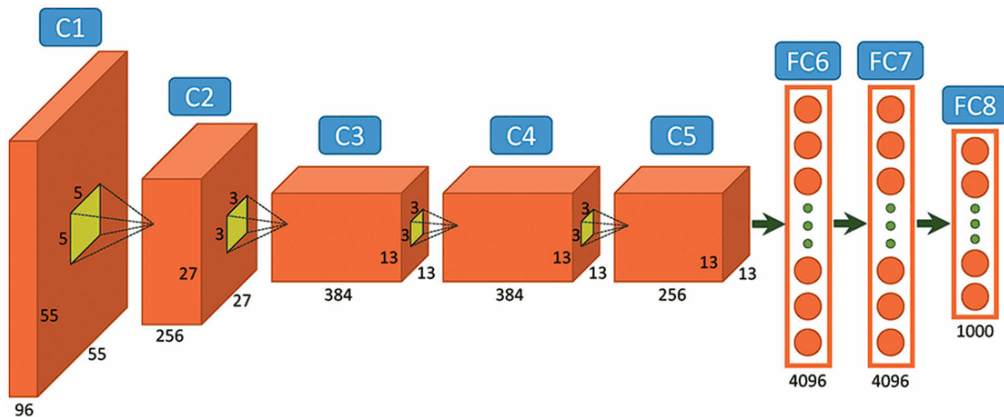
Quantitative Techniques

- Normal, Poisson, Binomial,
- P-value, E-value
- de Bruijn and overlap graphs
- Quality Values (Phred Scale)
- Full text indexing & BWT
- Seed & Extend
- Differential Expression
- Expectation Maximization
- PCA / t-SNE / UMAP
- One-hot encoding, tokens
- Logistic Regression, Random Forest
- Convolutional Neural Networks
- Transformers / Attention

What is the goal? What is the approach? What are the key challenges?

How did we explore these topics in the homeworks and lectures?

Sample Question



Consider this block diagram of the AlexNet architecture

- Label the main component that assigns a probability to each output class
- Label the main component that can detect vertical lines
- Label the main component that can detect faces in an image
- Label the main component that can be used as a high dimensional embedding
- Now briefly describe how these components are used to classify an image
- Now discuss how this model is used for DeepVariant
- Now discuss how this model is modified for Enformer

Sample Question

For each of these assays, describe what it is used for, how it works, and the major experimental and computational factors involved

- a. RNA-seq
- b. scRNA-seq
- c. Methyl-seq
- d. Chip-seq
- e. Hi-C